

• P_1 : $\ell \equiv 1 \pmod{3}$, undetected
 ▼ R : detects χ , but β -summable



10^1

10^2

10^3

ℓ (log scale)

density of $\Theta_S \Leftrightarrow R \neq \emptyset$ (existence)
 uniqueness $\Leftrightarrow \sum_{\ell \in R} \ell^{-\beta} = \infty$ (abundance)